

LUGOJ, Romania

WMO: 152700

Lat: 45.6867N

Lon: 21.9333E

Elev: 124

StdP: 99.84

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-10.3	-7.4	-12.4	1.3	-8.7	-9.6	1.7	-6.4	10.3	3.3	9.5	2.4	1.2	120	0.449

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	12.9	33.6	21.6	31.9	21.5	30.1	21.0	23.4	29.4	22.6	28.9	21.8	28.2	2.4	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.7	16.7	25.8	20.8	15.7	24.9	20.0	14.9	23.9	70.3	29.5	67.2	28.7	64.2	28.4	26.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	8.4	7.0	5.6	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)				N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	12.0	0.2	3.7	7.1	12.0	16.9	20.6	21.7	22.3	17.3	12.1	7.6	2.3	(6)
		DBStd	8.54	4.84	4.42	4.10	4.71	3.32	3.23	3.02	3.21	4.25	4.46	4.72	4.29	
		HDD10.0	1000	304	181	108	34	1	0	0	1	32	100	240		
		HDD18.3	2693	562	409	349	192	67	16	6	5	70	197	322	497	
		CDD10.0	1737	1	4	16	95	213	318	363	381	219	96	28	2	
		CDD18.3	389	0	0	0	3	22	84	112	127	38	3	0	0	
		CDH23.3	4078	0	0	2	67	250	830	1159	1337	378	56	1	0	
		CDH26.7	1538	0	0	0	6	46	283	462	609	126	6	0	0	
(14)	Wind	WSAvg	2.0	1.9	2.4	2.2	2.2	2.0	1.8	1.8	1.7	1.7	1.7	2.4	1.9	(14)
(15)	Precipitation	PrecAvg	710	44	39	40	61	72	89	78	57	68	59	51	57	(15)
(16)		PrecMax	969	86	121	117	105	172	203	167	116	186	132	127	146	(16)
(17)		PrecMin	463	13	2	7	13	28	29	11	8	12	2	0	7	(17)
(18)		PrecStd	134	21	25	27	23	35	43	42	29	36	34	29	34	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	DB	13.5	16.6	22.0	27.4	29.8	33.3	35.4	36.2	33.3	27.4	20.9	14.7		(19)
(20)		MCWB	9.6	11.0	13.2	17.0	19.0	23.0	21.5	21.5	20.5	17.4	13.8	10.2		(20)
(21)		DB	10.0	14.3	19.0	25.4	28.2	31.6	33.4	34.0	29.7	24.7	18.2	11.7		(21)
(22)		MCWB	7.4	9.7	11.4	16.2	18.9	22.5	21.5	21.4	19.3	16.3	13.2	9.0		(22)
(23)		DB	8.0	12.4	16.2	22.9	26.2	29.9	31.1	32.2	27.5	22.3	16.6	9.8		(23)
(24)		MCWB	6.4	8.3	10.2	14.5	17.9	21.6	21.3	21.2	18.6	15.6	12.4	7.8		(24)
(25)		DB	6.4	10.3	13.8	20.7	24.4	28.2	29.1	30.0	25.1	19.6	14.5	7.6		(25)
(26)		MCWB	5.0	7.4	8.6	13.5	17.1	21.0	20.6	20.8	18.1	14.6	11.5	6.3		(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	WB	10.1	11.9	13.9	18.0	21.2	25.2	23.9	24.2	21.5	19.0	14.9	11.2		(27)
(28)		MCDB	13.2	16.2	21.6	25.4	27.7	31.0	29.2	31.4	30.8	24.9	18.3	13.1		(28)
(29)		WB	8.0	10.2	12.0	16.8	20.0	23.7	23.0	23.0	20.3	17.3	13.8	9.4		(29)
(30)		MCDB	9.5	13.5	17.5	24.1	25.8	29.1	28.8	29.9	27.3	22.7	17.8	12.0		(30)
(31)		WB	6.7	8.8	10.7	15.6	19.0	22.7	22.2	22.2	19.6	16.2	12.9	7.8		(31)
(32)		MCDB	7.9	11.5	15.3	22.3	24.4	28.5	28.8	29.0	25.4	20.9	16.3	9.5		(32)
(33)		WB	4.8	7.7	9.5	14.1	17.9	21.6	21.4	21.5	18.7	15.1	11.8	6.6		(33)
(34)		MCDB	6.3	10.0	13.1	19.2	22.6	26.9	27.8	28.1	23.7	19.2	14.4	7.8		(34)
(35)	Mean Daily Temperature Range	MDBR	6.4	8.2	10.4	11.6	11.4	11.9	12.5	12.9	11.2	11.3	8.1	6.2		(35)
(36)		5% DB	8.6	11.2	14.3	15.4	14.8	14.6	15.7	16.2	15.1	15.7	11.1	9.7		(36)
(37)		MCWBR	6.1	7.0	7.9	7.3	6.6	6.2	5.8	5.1	5.8	7.8	6.4	6.5		(37)
(38)		5% WB	8.0	9.5	12.3	13.7	12.9	13.2	13.2	14.0	12.8	14.0	9.9	9.2		(38)
(39)		MCWBR	5.9	6.3	7.1	6.7	6.5	6.4	5.6	5.4	5.4	7.3	5.8	6.4		(39)
(40)	Clear-Sky Solar Irradiance	taub	0.334	0.362	0.402	0.459	0.445	0.437	0.451	0.461	0.429	0.388	0.356	0.334		(40)
(41)		taud	2.332	2.253	2.199	2.059	2.160	2.225	2.191	2.160	2.214	2.324	2.364	2.358		(41)
(42)		Ebn at Noon	753	799	817	800	826	834	816	790	780	762	721	710		(42)
(43)		Edh at Noon	78	102	124	156	146	137	140	140	122	95	76	70		(43)
(44)	All-Sky Solar Radiation	RadAvg	1.23	2.05	3.25	4.50	5.41	6.01	5.96	5.37	3.78	2.61	1.54	0.99		(44)
(45)		RadStd	0.14	0.27	0.43	0.46	0.49	0.53	0.48	0.51	0.41	0.28	0.15	0.14		(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (65 neighbors)	+0.72	N/A	N/A	N/A	N/A	N/A	-131	-198	+128	+64

Nomenclature: See separate page